Name: _____ Roll No.: _____

EE 2001 Digital Systems Quiz II

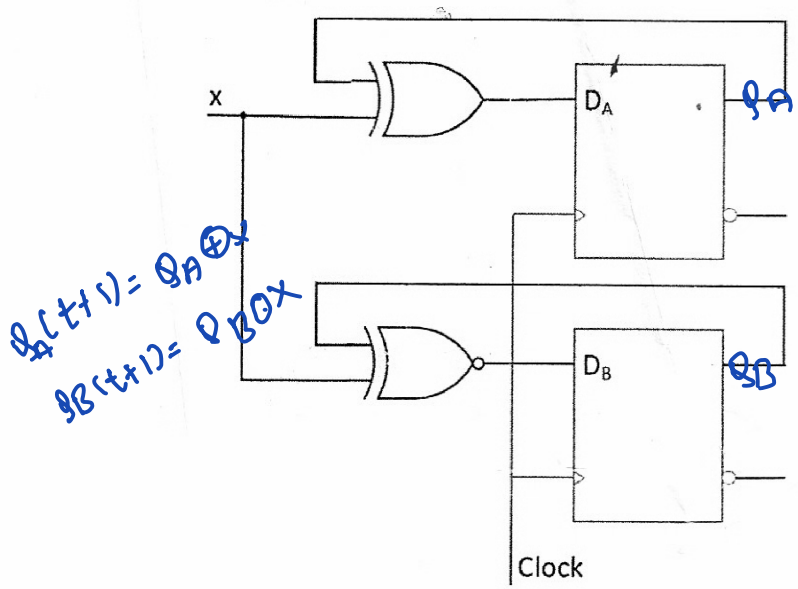
Monday, 18/03/2024

Closed book, 50 mins, 20 points

Remember...

- *Make reasonable assumptions when in doubt. However, you should show all steps and justify assumptions for full credit.*

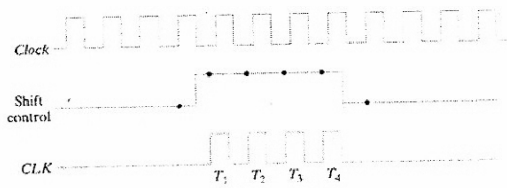
1. Analyze the synchronous sequential circuit provided below and determine the functionality realized by this circuit. (5 pts)

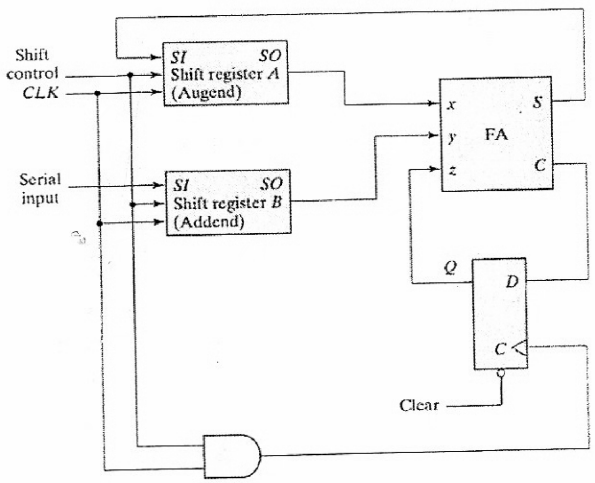


2. Design a mod-5 synchronous counter and realize the same using T flip flops. Make sure that any unused state occurrence leads the circuit to return to the initial state i.e., all zeros. (5 pts)

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3. (a) The serial adder of the below figure uses two four-bit registers. Register A holds the binary number 1010 and register B holds 0100. The carry flip-flop is initially reset to 0. List the binary values in register A and the carry flip-flop after every clock cycle. (2 pts)





(b) Design a traffic light controller using a 3-bit ring counter such that RED, YELLOW, GREEN lights turn ON sequentially with every clock cycle. (3 pts)

4. Consider the circuit below. Assume initial states as all 0's. Find the count sequence (Q_A, Q_B, Q_C) and Mod number of the given counter. If we want to reverse the count sequence, what changes are needed in the circuit? (5 pts)

